ITAI 4374

Module 07

Attention and Consciousness in AI

How does the brain choose what matters — and could a machine ever truly be aware?

Houston City College • AI & Robotics Program • Spring 2026

Where We Are in the Journey

✓	Module 02: COMPUTE	What does the brain compute?
✓	Module 03: BUILD	Brain anatomy & neurons
✓	Module 04: CONNECT	Spiking neural networks
✓	Module 05: SENSE	Sensory processing & perception
✓	Module 06: REMEMBER	Memory & learning
▶	**Module 07: ATTEND & AWARE**	Attention & consciousness

The Problem: Too Much Information

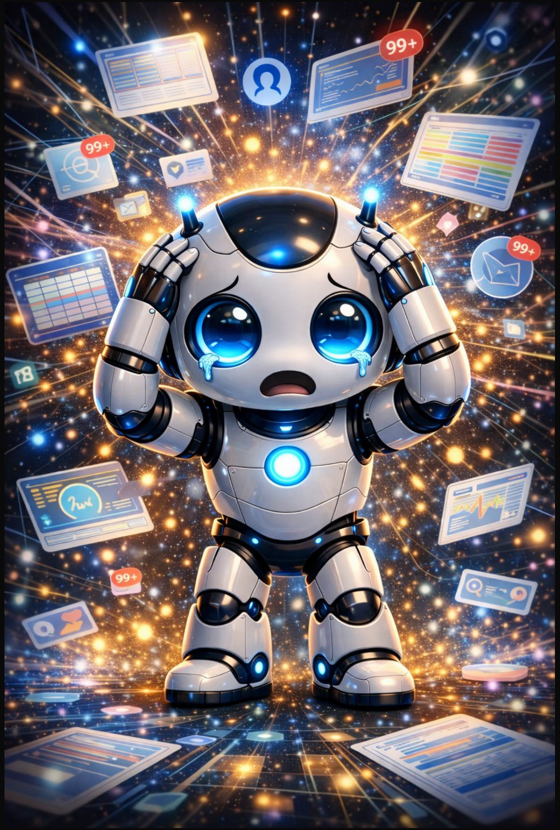
Your brain receives ~11 million bits of sensory information every second.

Your conscious mind can process about 50 bits per second.

That means you are consciously aware of less than 0.0005% of incoming data.

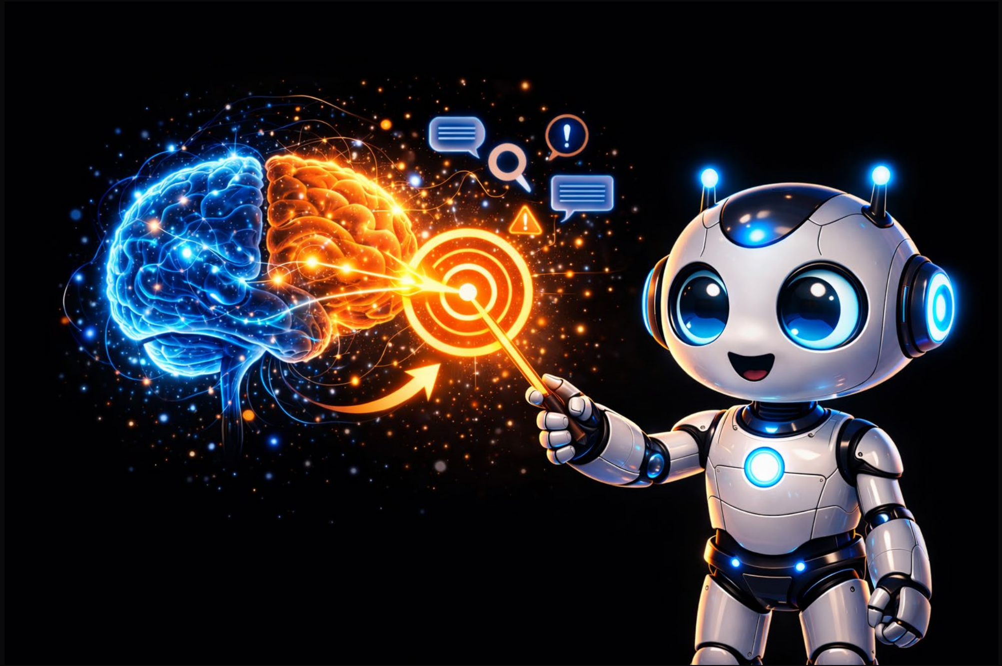
The brain needs a filter — a bouncer at the door of consciousness.

That filter is ATTENTION.



PART 1

How the Brain Pays Attention



The Cocktail Party Effect

At a crowded party, you can follow one conversation despite noise everywhere.

But if someone across the room says YOUR NAME — you hear it.

Your brain was monitoring background noise without you knowing.

Attention is not just about focusing on one thing.

It operates on multiple levels simultaneously.

— Colin Cherry, 1953

Three Types of Attention

Sustained

Staying focused over long periods

Security guard watching monitors

Selective

Focusing on one thing, ignoring rest

Studying in a noisy coffee shop

Divided

Attempting to focus on multiple things

Texting while driving (badly!)

Misconception Alert: "I'm a Great Multitasker"

Research consistently shows: almost no one truly multitasks.

"Multitasking" is actually rapid task-switching with a measurable performance cost.

People who believe they are good at multitasking actually perform WORSE on attention tasks.

The only real multitasking: combining an automatic skill (walking) with a conscious one (talking).

Every switch has a cost. Every time.

Inattentional Blindness: The Invisible Gorilla

Simons & Chabris (1999):

Inattentional Blindness: The Invisible Gorilla

Participants counted basketball passes in a video.

A person in a GORILLA SUIT walked through the scene. On screen for 9 full seconds. Beat their chest.

~50% of participants did not see the gorilla.

Their eyes received the image. Their visual cortex processed it.

But their ATTENTION was elsewhere — so the gorilla never reached consciousness.

Seeing ≠ Noticing

Two Modes of Attention

BOTTOM-UP

Stimulus-driven
Automatic & involuntary

A loud bang
A flash of light
A spider on your arm

"Something GRABBED me"

TOP-DOWN

Goal-driven
Voluntary & deliberate

Looking for your car
Listening for your flight
Reading this slide

"I CHOSE to focus"

The Brain's Three Attention Networks

Michael Posner & colleagues identified three distinct networks,

each with different brain regions and different jobs.

Think of them as three departments in a company:

1. ALERTING — "Wake up, something might happen"

2. ORIENTING — "Look over there"

3. EXECUTIVE — "Focus on this, ignore that"

Alerting Network

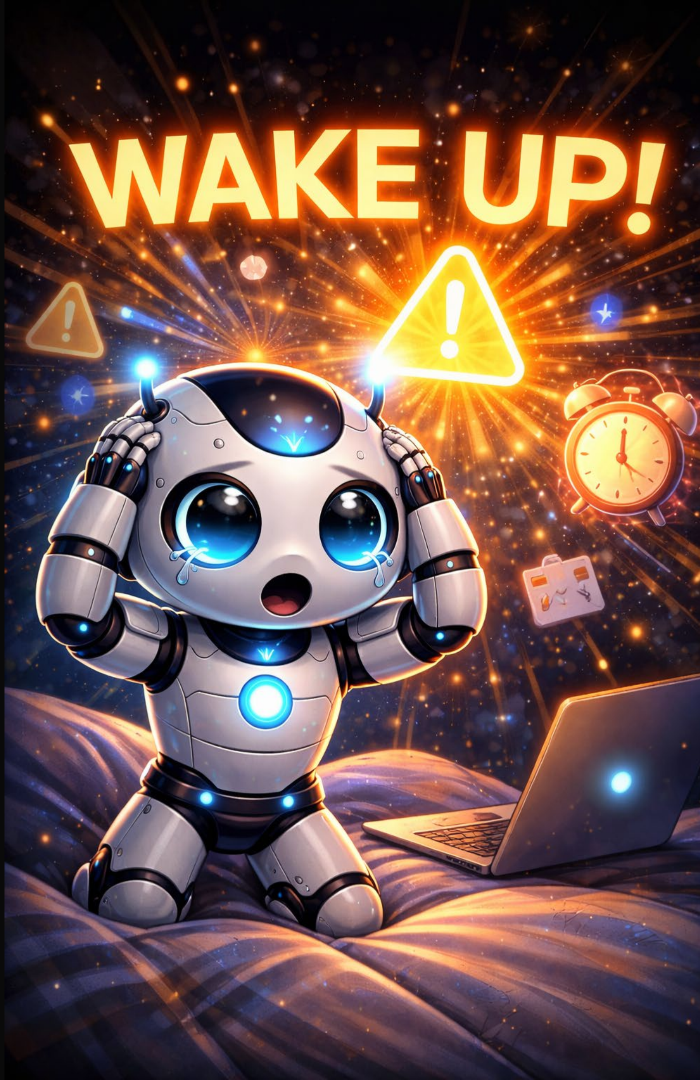
"Wake up — something might happen"

The brain's alarm system

Brain regions: Brainstem (locus coeruleus) + Thalamus
Neurotransmitter: Norepinephrine

Doesn't tell you WHAT to focus on —
just puts the whole brain on high alert.

Analogy: A security alarm goes off. It says "something is happening" but not which door opened.



Orienting Network

"Look over there"

The brain's GPS for attention

Brain regions: Parietal cortex (right side) + Superior colliculus

Shifts your attention spotlight from one place to another.

Analogy: Someone calls your name — your orienting network swivels your attention to the source.

Executive Network

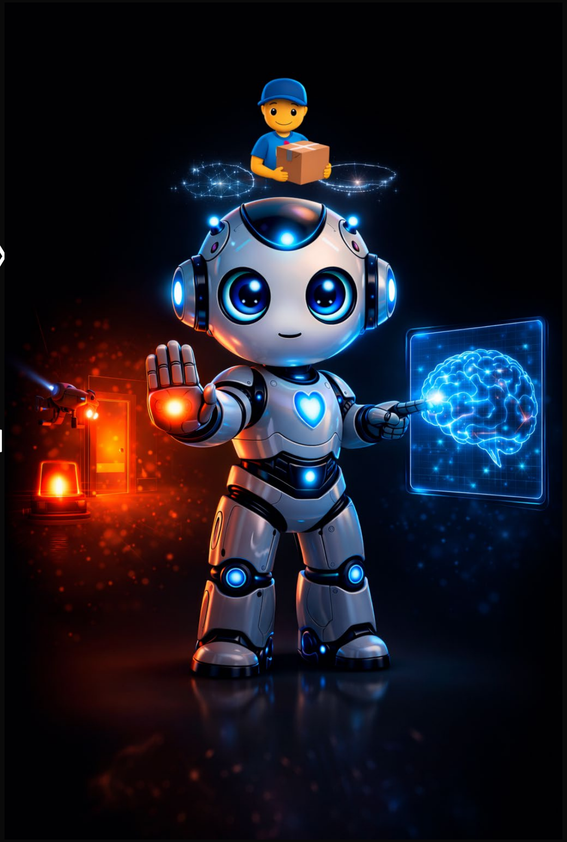
"Focus on this, ignore that"

The brain's CEO

Brain regions: Prefrontal cortex (PFC) + Anterior cingulate cortex (ACC)

- Handles conflicts, overrides impulses, directs voluntary focu
- Not fully mature until mid-20s (why teens struggle with impulse control).

Analogy: The alarm went off (alerting), security pointed at the north door (orienting), but the CEO says: *"Just the delivery person. Keep working."*



The Thalamus: Brain's Gatekeeper

From Module 05: all sensory info (except smell) passes through the thalamus on its way to the cortex.

- But it's not just a relay. It's a GATEKEEPER.

The PFC sends signals DOWN to the thalamus:

- *"Turn UP visual info. Turn DOWN sounds. I'm reading."*

Critical insight:

- Attention is not just amplifying what you want.
- **It is equally about SUPPRESSING what you don't want.**

Spatial vs. Feature vs. Object Attention

SPATIAL

Focus on a LOCATION

"Spotlight" model
Narrow & intense or
wide & diffuse

FEATURE

Focus on a
PROPERTY
across entire field

Why red traffic lights
"pop out" anywhere

OBJECT

Focus on an entire
OBJECT as one unit

Binds features together
(handle + color +
shape)

The Biased Competition Model

Desimone & Duncan (1995):
Every object, feature, and location is represented by neurons.
These neural groups COMPETE for processing resources.

Attention is the bias that tips the election.

Top-down or bottom-up signals give one group "extra votes."
Winners get amplified. Losers get suppressed.

AI parallel: This is what softmax does in transformers —
a competition where winners get highest weights.

When Attention Breaks: ADHD

Misconception: "ADHD means you can't pay attention."

Reality: ADHD is a disorder of attention REGULATION, not attention CAPACITY.

Someone with ADHD can focus for hours on a video game
(constant stimulation, immediate rewards)...
...but struggles with a textbook for 10 minutes
(requires sustained top-down control, no immediate reward).

The executive network struggles to override
the pull of more interesting stimuli.

When Attention Breaks: Hemispatial Neglect

Damage to right parietal cortex (orienting network):

- Patients literally IGNORE the left side of their world.
- Eat from right side of plate only. Shave right side of face.
- Draw clock with all 12 numbers crammed on the right.

They are NOT blind on the left side.
Their visual system is intact.

- The attention system no longer directs resources leftward.
- Without the spotlight, the left side ceases to exist for awareness.

Perception REQUIRES attention.

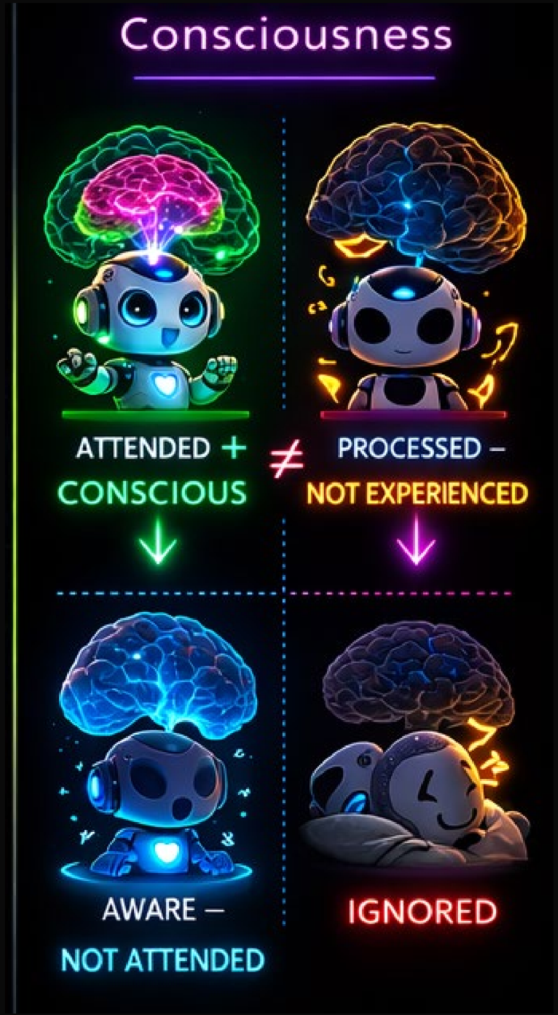
Why This Matters for AI

Clinical cases reveal: attention is NOT optional.

Even with all the right sensory hardware, without effective attention, a system cannot function in the real world.

Early AI without attention = like Balint syndrome
(could only see one object at a time, overwhelmed by data)

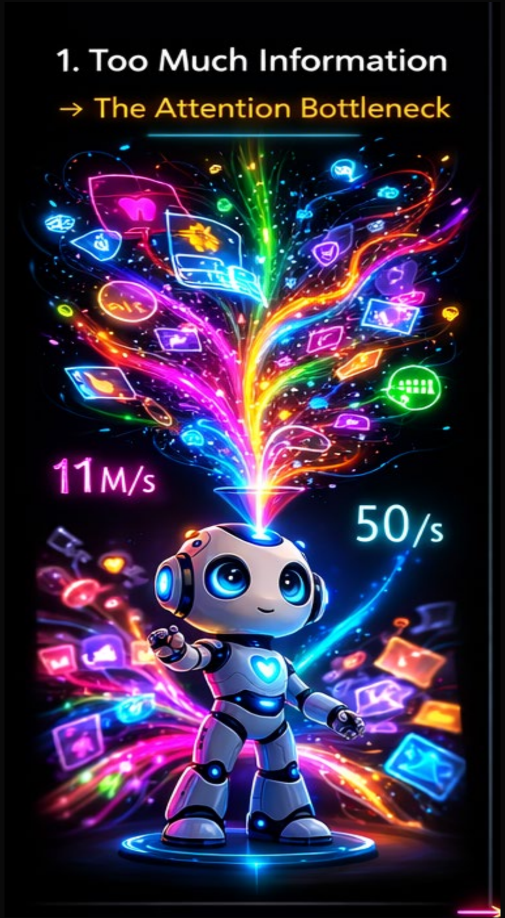
Adding attention to AI was not just an optimization —
it was a fundamental requirement for real-world AI.



PART 2

AI Attention & The Consciousness Question

The AI Bottleneck Problem



Before attention (early seq-to-seq models):

Encoder reads entire sentence → compresses to ONE vector → Decoder

Like remembering a paragraph by condensing it to one sentence.

Works for short inputs. Falls apart for long ones.

The single vector can't hold enough information.

Same problem the brain faces:

too much input, too little processing capacity.

Same solution: ATTENTION.

Bahdanau Attention (2014): Learning to Look Back

Instead of one compressed vector, the system can:
"look back" at the original input at each step.

For each output word, it computes:

"Which input words are most relevant right now?“

This is the AI version of the ORIENTING NETWORK:
"Where should I focus at this moment?"

Result: Dramatic improvement, especially for long sentenc

Self-Attention & The Transformer (2017)

Self-attention: every element checks its relationship with every other element.

"The cat sat on the mat because it was tired"

→ *Self-attention figures out "it" = "the cat"*

"Attention Is All You Need" (Vaswani et al., 2017)

Multi-head attention = multiple spotlights in parallel, each focusing on different aspects (grammar, meaning, position).

Like the brain running spatial + feature + object attention simultaneously through different circuits.

Brain ↔ AI Attention Mapping

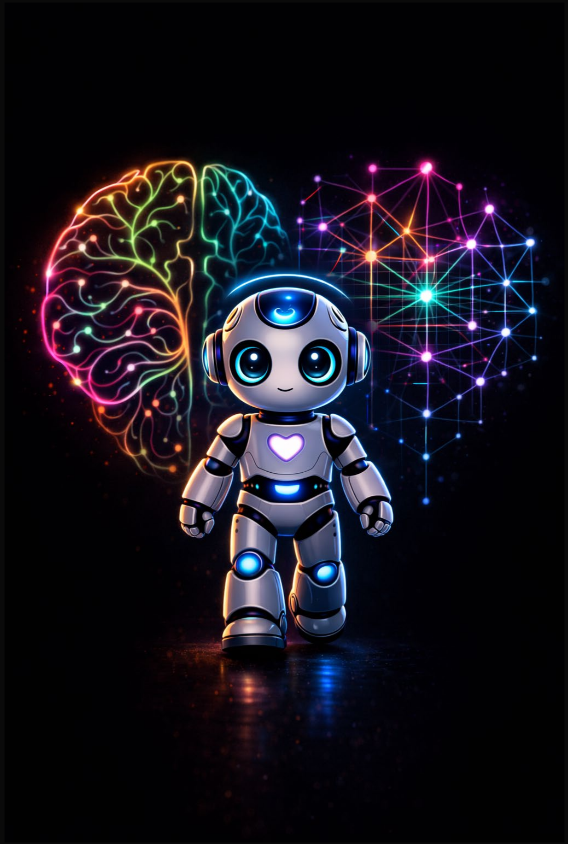
Brain Mechanism	AI Equivalent
Bottom-up attention (stimulus grabs you)	High attention weights on surprising tokens
Top-down attention (you choose to focus)	Query vectors that specify what the model looks for
Orienting network (shift focus to location)	Positional encoding + attention on positions
Executive network (resolve conflicts)	Softmax competition: only most relevant win
Multi-system attention (spatial + feature + object)	Multi-head attention (parallel patterns)

Where Bio and AI Attention Differ

Despite the shared name and inspiration, biological and AI attention are fundamentally different.

Understanding these gaps tells us:

- Where AI has room to improve
- Where the brain is still far ahead
- What "attention" really means in each context



Biological vs. AI Attention

Dimension	Brain	AI
Energy	~20 watts (whole brain)	Thousands of watts
Adaptability	Adapts instantly	Fixed after training
Emotion	Amygdala modulates priority	No emotional component
Context	Unlimited (via LTM)	Fixed window (e.g., 128K)
Learning	Single experience	Massive training data
Embodiment	Eyes, head, posture	Purely computational
Default state	Default Mode Network	No equivalent — just off
Consciousness	Linked to awareness	No evidence of awareness

The Emotion Gap

In the brain, the amygdala acts as an emotional early-warning system.

It can HIJACK attention:

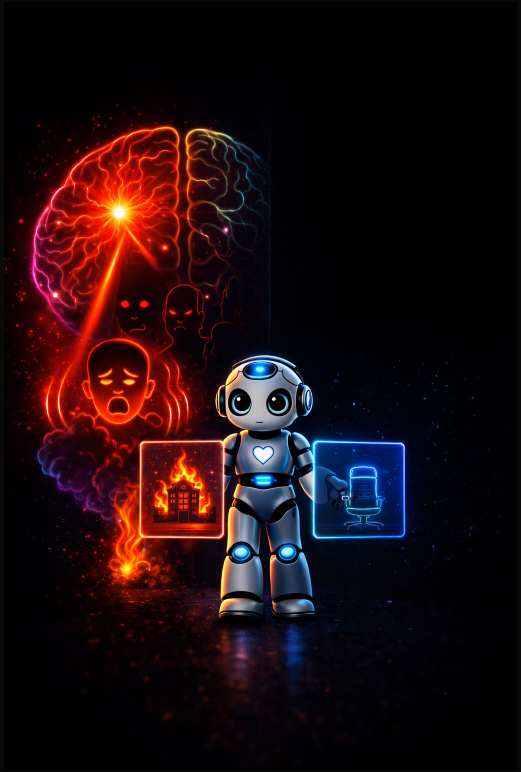
Threatening face in a crowd → immediate priority

Crying child → can't ignore it

Smell of smoke → overrides everything

Current AI treats "the building is on fire" and "fire the CEO" with the same computational indifference.

Could RLHF be a crude form of emotional modulation? (A feedback signal that biases behavior, like dopamine...)



The Default Mode Network: When You Stop Paying Attention

When not focused on a task, the brain doesn't just idle.

The Default Mode Network (DMN) activates:

- Daydreaming and mind-wandering
- Self-reflection
- Thinking about the future
- Imagining others' perspectives
- Creative incubation

Why you solve problems in the shower!

AI has nothing like this. When not processing a query, it is simply OFF.

Now for the Big Question...

What is consciousness?

And could a machine ever have it?

What Is Consciousness?

The difference between being AWAKE and being AWARE.

The difference between PROCESSING information and EXPERIENCING it.

When you see red, your brain processes a wavelength of light.

But you also EXPERIENCE redness.

A subjective, first-person sensation.

Where does that come from?

Four major theories attempt to answer this.

Theory 1: Global Workspace Theory

Baars (1988), expanded by Dehaene

Consciousness as a THEATER:

Backstage = unconscious processing (breathing, balance, etc.)
Stage = the global workspace (limited capacity)
Spotlight = attention (selects what reaches the stage)
Audience = all brain regions receiving the broadcast

Information becomes CONSCIOUS when it wins the **competition and gets BROADCAST widely across the brain.**

AI parallel: Global Workspace architectures (Goyal & Bengio, 2022) create shared workspaces where modules compete for broadcast.

Theory 2: Integrated Information Theory (IIT)

Tononi (2004)

Consciousness = INTEGRATED information

(information that can't be broken into independent parts)

Measured by Phi (Φ) — higher Phi = more conscious

A photograph: millions of pixels, but each is INDEPENDENT.
Low integration. Low Phi.

Your experience of a sunset: colors, warmth, emotion, memory .
All woven into ONE indivisible experience. High Phi.

Bold prediction: any system with high enough Phi is conscious, regardless of what it's made of. But Phi is impossible to calculate.

Theory 3: Higher-Order Theories (HOT)

Rosenthal, Lau, and others

Consciousness = having THOUGHTS about your thoughts

It's not enough to see red (first-order).

Consciousness requires KNOWING you see red (higher-order).

A thermostat detects temperature (first-order).

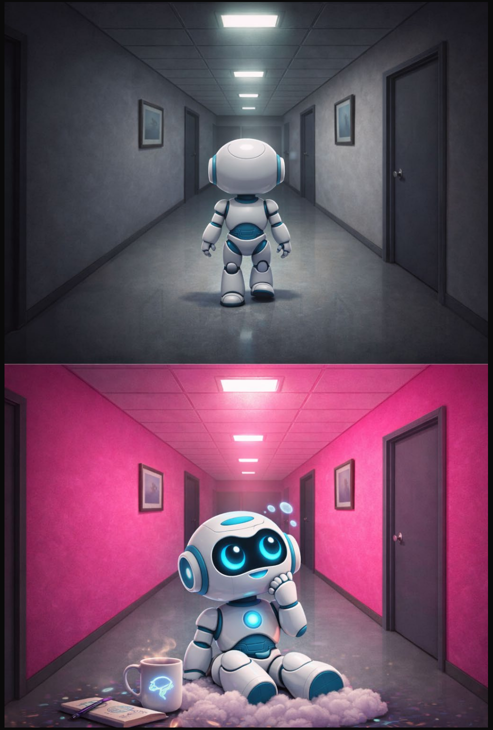
It doesn't KNOW it detects temperature.

AI has meta-learning and self-monitoring...

but is it EXPERIENCING those meta-representations?

The computation is there. The experience... uncertain.

Theory 4: Predictive Processing



Friston, Clark, and others

Remember Module 02? The brain is a prediction machine.

- Predictions match reality → smooth, largely unconscious.
- Prediction MISMATCH → you become CONSCIOUS of it.
- Walking down a familiar hallway: barely aware of it.
- *Wall painted bright pink overnight: very conscious of it!*
- Attention controls how much weight goes to prediction errors.
- High attention = errors amplified = more conscious awareness.
- **Ties attention and consciousness together elegantly.**

Four Theories — Each Captures Something True

Theory	Key Claim	Explains Why...
GWT	Consciousness = broadcast	...awareness feels like suddenly "knowing"
IIT	Consciousness = integration	...experience feels unified, not fragmented
HOT	Consciousness = meta-thought	...we can reflect on our own awareness
Pred. Proc.	Consciousness = prediction error	...novelty and surprise feel so vivid

No single theory has won. The final answer may incorporate all four.

The Hard Problem of Consciousness

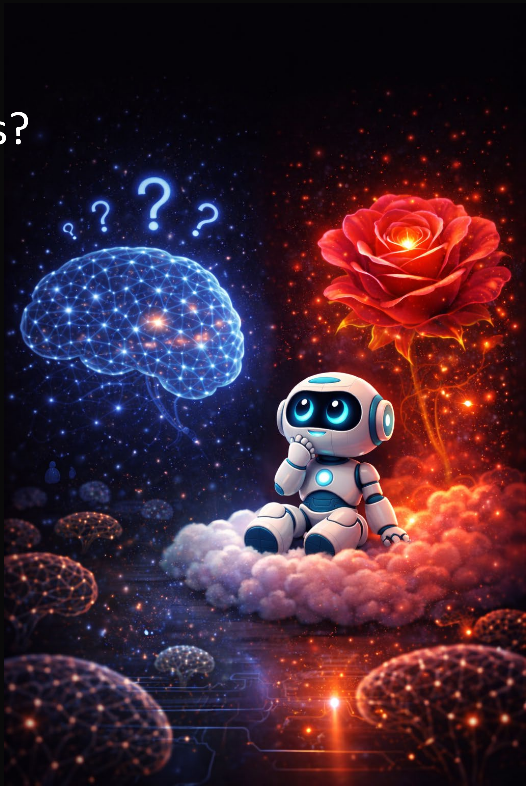
David Chalmers, 1995

Easy problems (hard, but solvable in principle):
How does the brain integrate info? Direct attention? Report states?

THE HARD PROBLEM:

Why is there SUBJECTIVE EXPERIENCE at all?

Why does seeing red FEEL like something?
Why doesn't all this neural activity happen
"in the dark" without any felt quality?



The Chinese Room (Searle, 1980)

You're locked in a room with a rulebook (in English). People slide in Chinese characters. You follow rules, slide back Chinese responses.

Outside: it looks like the room understands Chinese.

Inside: YOU understand NOTHING. Just following rules.

Searle's argument: This is what a computer does. Manipulating symbols without understanding meaning.

Counterargument (Systems Reply):

No single neuron understands English either.
Understanding is a property of the WHOLE SYSTEM.

Where Current AI Stands

The honest assessment:

There is NO evidence any current AI system is conscious.

> LLMs can claim to have experiences, express emotions, and argue they are conscious.
>
> But producing language ABOUT consciousness **is not the same as BEING conscious.**

Current AI lacks: persistent self, embodiment, emotional states, continuous experience, unified perspective.

But we also don't have a reliable TEST for consciousness.
We assume humans are conscious because they're like us.

Attention ≠ Consciousness

ATTENDED \ CONSCIOUS	YES	NO
YES	Normal Perception Focused reading, active listening	Subliminal Priming Blindsight Brain processes it but you don't experience it
NO	Peripheral Awareness? (debated) Aware of room temp without focusing on it	Truly Unconscious Digestion, immune response, etc.

Why This Matters: Real-World Implications

Safety: If AI could suffer, we have moral obligations. Almost certainly not yet — but as systems become more brain-like, the question grows.

Ethics: If AI mimics consciousness convincingly but isn't conscious, could humans form false attachments?

Legal: If AI were conscious, would it have rights? Legal scholars are already debating this.

Design: We should be careful about systems that claim or imply consciousness when we cannot verify it.

Key Takeaways

ATTENTION is well understood.

We know the brain regions, the mechanisms, the clinical cases, and the AI implementations.

CONSCIOUSNESS remains one of the deepest mysteries in science.

The two are RELATED but SEPARABLE.

Attention is necessary but NOT sufficient for consciousness.

As AI engineers, you need to understand both —
the mechanisms we CAN build and the mysteries we CANNOT yet solve.

Lab Preview: Attention in Brains & Machines

Part A: Human Attention Experiments

Stroop test, visual search, change detection
Record YOUR performance data

Part B: AI Attention Visualization

Visualize transformer attention maps
What does the model attend to?
(No GPU required — pre-trained model, CPU only)

Part C: Comparison & Reflection

Write a structured comparison of bio vs. AI attention
Connect to the theories from this module

Discussion Questions

1. If a future AI system passed every behavioral test for consciousness, should we treat it as conscious? Why or why not?
2. Social media is designed to exploit bottom-up attention. Should there be regulations on attention-hijacking design?
3. Which consciousness theory do YOU find most convincing, and what does it imply for the possibility of machine consciousness?
4. Is the Chinese Room argument still relevant in the age of LLMs?

Next: Module 08 — AI and Human Decision-Making

Module 07 asked: How does the brain choose what to NOTICE?

Module 08 asks: How does the brain choose what to DO?

Coming up:

- Dopamine reward system & the orbitofrontal cortex
- Cognitive biases & why humans decide irrationally
- Reinforcement learning ↔ brain reward circuits
- The alignment problem: making AI decisions match human values
- Case studies: when AI decisions went wrong